Vimansh Mahajan

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LinkedIn | Portfolio | GitHub

Education

Indraprastha Institute of Information Technology, Delhi (IIIT-Delhi) CGPA: 9.28 / 10.00
Bachelor of Technology (B.Tech.) in Computer Science Engineering (CSE) *2022 – 2026*
• Dean's List (2024–25) • Flipkart GRiD 7.0 National Semi-Finalist • Adobe GenSolve Top 5%

The Vivekanand School, Narela, New Delhi CBSE Class XII: 97.2%
• JEE Main 2022: 99 Percentile (AIR 9,200) • JEE Advanced 2022: AIR 10,109

Experience

Bank of New York (BNY) Aug 2026 – Present
Engineering Analyst – Software Engineering
- Developed UI components, performed end-to-end testing and bug fixes for service deployments, and supported maintenance of the **Structured Debt Platform**.
- Contributed to **Spring Boot 3.5 to 4.1 upgrades** for Spring Cloud services and led a team knowledge-sharing session on the upgrade process; received a **VIP Recognition Award** and assumed **pod-level Release Coordinator** responsibilities.

Bank of New York (BNY) May 2025 – July 2025
Software Engineer Intern
- Architected a production-grade content automation system using **Spring Boot**, **Python LLM services**, **Oracle DB**, and **Angular**, transforming raw finance notes and images into publication-ready market commentary.
- Reduced content turnaround time from **4–5 hours to under 10 minutes** by automating structured note capture, AI drafting in author style, contextual image insertion, and final publishing workflows.
- Containerized backend services and deployed production workloads on **Google App Engine**, improving portability and release efficiency.

Complex Systems (CoSy) Lab Jan 2025 – May 2025
UG Researcher
- Developed RecipeGPT on the **Recipe1M corpus**, achieving BERTScore above 0.85 and ROUGE-L of 0.32 for recipe generation.
- Built an NER-driven preprocessing pipeline and a **three-stage GPT-2 fine-tuning workflow** over 100K+ recipe instructions, incorporating ingredient-title conditioning and label masking.

Projects

OptiWealth: Portfolio Optimization Platform | Live Demo | *Spring Boot, Python, React, PostgreSQL* Nov 2025
- Architected and deployed a multi-service portfolio analytics platform, integrating a **React** frontend, **Spring Boot REST APIs**, and Python-based analytics services.
- Secured user access with **JWT authentication** and Spring Security; built analytics for NSE-listed equities, including returns, volatility, Sharpe ratio, VaR/CVaR, beta, drawdown, and diversification.
- Built forecasting and portfolio optimization workflows using **ARIMA**, **GARCH**, Monte Carlo simulation, and efficient-frontier strategies; integrated AI-generated investment insights and automated deployment through **GitHub Actions**.

OncoHelp-AI: Brain MRI Decision Support | *Python, Streamlit, Grad-CAM, LLMs* Dec 2025
- Architected a multi-stage **brain MRI decision-support system** using ResNet18 and EfficientNet models for image validation, tumor screening, and subtype classification.
- Integrated **Grad-CAM heatmaps** and **Monte Carlo Dropout**–based uncertainty estimation to surface interpretable predictions with calibrated confidence signals.
- Developed a safety-constrained LLM assistant using Gemini and BioGPT for conservative summaries and follow-ups.

Skills

Languages: Python, Java, C++
Frameworks & Libraries: Spring Boot, React, Angular, Hugging Face Transformers, PyTorch
Databases & Tools: PostgreSQL, MySQL, Oracle DB, Git, Docker, REST APIs
Core CS: Data Structures & Algorithms, OOP, DBMS, Computer Networks, System Design
AI / ML: Machine Learning, NLP, LLM Fine-Tuning, Model Evaluation, Data Science

Additional Information

Coding Platforms: Leetcode (300+ problems solved), Codeforces
Certification: Quantitative Finance & Algorithmic Trading in Python
Volunteering: Volunteer, Seva Sankalp NGO